DAA LAB — WEEK 1
Programs 1–5
Source: extracted from your uploaded DAA lab PDF.

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PROGRAM 1: Counter method: triangular loop
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```
#include<stdio.h>
int sum(int n)
{
int i= 0;
   int s =0;
   while(s <= n)
   {
        i++;
        s += i;
    }
return i*3;
}
int main()
{

int n;
scanf("%d",&n);
printf("%d",sum(n));
}
```

==
PROGRAM 2: Counter method: nested loops with break
==

```
#include<stdio.h>
void func(int n)
{
   if(n==1)
   {
     //printf("*");
   }
   else
   {
    for(int i=1; i<=n; i++)
    {
      for(int j=1; j<=n; j++)
      {
         //printf("*");
         //printf("*");
         break;

      }
    }
  }
```

```
  printf("%d",(5*n+2));
}
int main()
{
    int n;
    scanf("%d",&n);
    func(n);
}
```

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PROGRAM 3: Counter method: factor of a number

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```
#include <stdio.h>
int main()
{
   int n, i, counter = 0;
   scanf("%d", &n);
   counter++;
   for (i = 1; i <= n; ++i)
   {
       counter++;
       if (n % i == 0)
       {
           counter++;
       }

       counter++;
   }
   printf("%d", counter);
   return 0;
}
```

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PROGRAM 4: Counter method: three nested loops

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```
#include <stdio.h>
int main()
{
   int n;
   int c = 0;
   int i, j, k;
   scanf("%d", &n);
   c++;
   for (i = n / 2; i < n; i++)

   {
       c++;
       for (j = 1; j < n; j = 2 * j)
       {
           c++;
```

```
            for (k = 1; k < n; k = k * 2)
            {
                c++;
                c++;
            }
            c++;
        }
        c++;
    }
    c++;
    printf("%d", c);
    return 0;
}
```

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PROGRAM 5: Counter method: reverse a number
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```
#include <stdio.h>
int main()

{
    int n;
    int rev = 0, remainder;
    int count = 0;
    scanf("%d", &n);
    count += 3;
    while (n != 0)
    {
        count++;
        remainder = n % 10;
        count++;
        rev = rev * 10 + remainder;
        count++;
        n /= 10;
        count++;
    }
    printf("%d", count);
    return 0;
}
```